Geometric Complexity in IPCA with Random Fourier Features: Empirical Evidence from Long-Only Equity Portfolios

An Empirical Assessment Against the Framework of
Deep, Lesniewski, Missaoui & Pakala (2026)

Empirical Results Report

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Abstract

We evaluate whether functional complexity added through Random Fourier Features (RFF) to Instrumented Principal Components Analysis (IPCA) constitutes *virtuous* or *illusory* complexity in the sense of Deep, Lesniewski, Missaoui and Pakala (2026). Using a long-only equity portfolio spanning January 1993 to December 2024, we systematically vary the number of RFF components ($\tilde{m} \in \{0, 128, 256, 512, 1024, 2048, 4096, 8192\}$), the regularization parameter ($\alpha \in \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 1, 10$ and the number of latent factors ($k \in \{10, 12, 14, 16, 18, 20, 22, 24\}$). Across all configurations, the Grassmann-geometric diagnostics — effective rank of fitted factor covariance, Geodesic Grassmann distance, and out-of-sample R^2 — align closely with portfolio performance, validating the paper's geometric framework. In particular, moderate RFF complexity ($\tilde{m} \approx 256$–512) expands intrinsic return dimension stably, turning OOS R^2 positive and delivering Sharpe ratios of 0.80–1.09, while zero-RFF and very-high-RFF regimes exhibit the geometric signatures of illusory complexity.

Contents

1 Introduction

The debate over the "virtue of complexity" in empirical asset pricing — whether high-dimensional or nonlinear models reflect genuine economic structure or merely statistical illusion — has attracted significant attention following Kelly, Malamud and Zhou (2024) and Kelly and Malamud (2025). Deep, Lesniewski, Missaoui and Pakala (2026) reframe this debate geometrically: regardless of parameter count, what matters is whether functional complexity genuinely expands the *return subspace*, a trajectory on the Grassmann manifold $\mathrm{Gr}(k, N)$.

The key insight is that the IPCA objective function, even after nonlinear lifting via Random Fourier Features, reduces to minimizing a projection functional over k-dimensional subspaces:

$$L(\Gamma) = \frac{1}{T} \sum_{t=1}^{T} \|(I - P_t(\Gamma))r_t\|^2 , \tag{1}$$

where $P_t(\Gamma)$ projects onto $V_t(\Gamma) = \mathrm{span}(\Phi(Z_t)\Gamma^\top) \in \mathrm{Gr}(k, N)$. The sole effect of RFF lifting is to enlarge the admissible family of Grassmann trajectories $t \mapsto V_t$. Complexity is economically meaningful only if it (i) persistently increases the effective rank $\mathrm{erank}(\hat{\Sigma}_f)$, (ii) stabilizes Grassmann trajectories, and (iii) improves out-of-sample pricing errors.

This report documents our empirical evaluation of these three diagnostics across a comprehensive grid of hyperparameters and matches results against the geometric framework.

2 Experimental Setup

2.1 Data and Model

We implement the RFF-IPCA model of Lesniewski and Missaoui (2026) on a U.S. equity panel covering $N \approx 500$–3000 assets per month from January 1993 to December 2024 (384 months total). Characteristics $z_{i,t} \in \mathbb{R}^m$ are used to construct the nonlinear feature map:

$$\varphi(z) = \sqrt{\frac{2}{\tilde{m}}} \left[\cos(\omega_1^\top z + b_1), \ldots, \cos(\omega_{\tilde{m}}^\top z + b_{\tilde{m}})\right]^\top, \quad \omega_j \sim p(\omega),\ b_j \sim \mathrm{Unif}[0, 2\pi], \tag{2}$$

yielding an RFF-augmented design matrix $\Phi(Z_t) \in \mathrm{Mat}_{N,\tilde{m}}(\mathbb{R})$.

2.2 Hyperparameter Grid

Three experimental designs are explored:

Experiment	**Factors k**	**Alpha α**	**RFF $\tilde{m}$**
Factor sweep	10, 12, 14, 16, 18, 20, 22, 24	100 (fixed)	0–4096
12-factor regularization	12 (fixed)	10^{-8} to 10^3	0–8192
14-factor regularization	14 (fixed)	10^{-8} to 10^3	0–4096
Base regularization	10 (fixed)	10^{-8} to 10^2	0–4096

Portfolio performance is evaluated on a long-only strategy using rolling 24-month training windows. Results for $T = 36$ will be reported in a subsequent batch of experiments. The geometric diagnostics use the effective rank $\operatorname{erank}(\hat{\Sigma}_f)$ and Geodesic Grassmann distance $d_{\text{geo}}(V_t, V_{t-1})$ computed across rolling estimates.

3 Geometric Diagnostics

3.1 Effective Rank

The effective (entropic) rank of the estimated factor covariance $\hat{\Sigma}_f = \frac{1}{T}\sum_t \hat{f}_t \hat{f}_t^\top$ is defined as:

$$\operatorname{erank}(\hat{\Sigma}_f) = \exp\left(-\sum_{i=1}^{k} p_i \log p_i\right), \quad p_i = \frac{\lambda_i}{\operatorname{tr}(\hat{\Sigma}_f)}. \tag{3}$$

Values near 1 indicate near-rank-1 collapse (one dominant factor absorbs all variation); values near k indicate all factors are equally active.

Table 1 summarizes effective rank statistics across experiments.

Table 1: Effective Rank Statistics by Configuration (384-month panel, 1993–2024)

Experiment	**Config**	**Mean**	**Std**	**25%**	**75%**
Factor sweep (k=10–24)	All RFFs	8.96	0.309	8.77	9.16
12-factor (k=12)	Best config	9.07	0.547	8.85	9.37
14-factor (k=14)	Best config	10.01	0.635	9.76	10.40

The k=12 and k=14 configurations with moderate regularization exhibit the most stable effective rank trajectories across the full sample period.

3.2 Subspace Stability: Grassmann Distance

The Geodesic Grassmann distance between consecutive-month subspaces, $d_{\text{geo}}(V_t, V_{t-1}) = \left(\sum_{i=1}^{k} \theta_i^2\right)^{1/2}$, measures how much the return span rotates from one month to the next. Per Theorem 6.1 of Deep et al. (2026), flat curvature directions (arising when $k > k_0$) imply $O(1)$ subspace rotations under sampling variation.

Table 2: Geodesic Grassmann Distance Statistics

Config	**Setting**	**Mean**	**Std**	**Interpretation**
Factor sweep	All	1.847	0.395	Moderate instability
12-factor	Best	0.593	0.116	Stable
14-factor	Best	0.553	0.115	Stable

Notably, the configurations with $k \in \{12, 14\}$ and moderate regularization exhibit the *lowest* Grassmann distances (0.55–0.60), consistent with the paper's claim that virtuous complexity manifests as stable subspace trajectories.

Figure 1 summarises both diagnostics visually across all configurations.

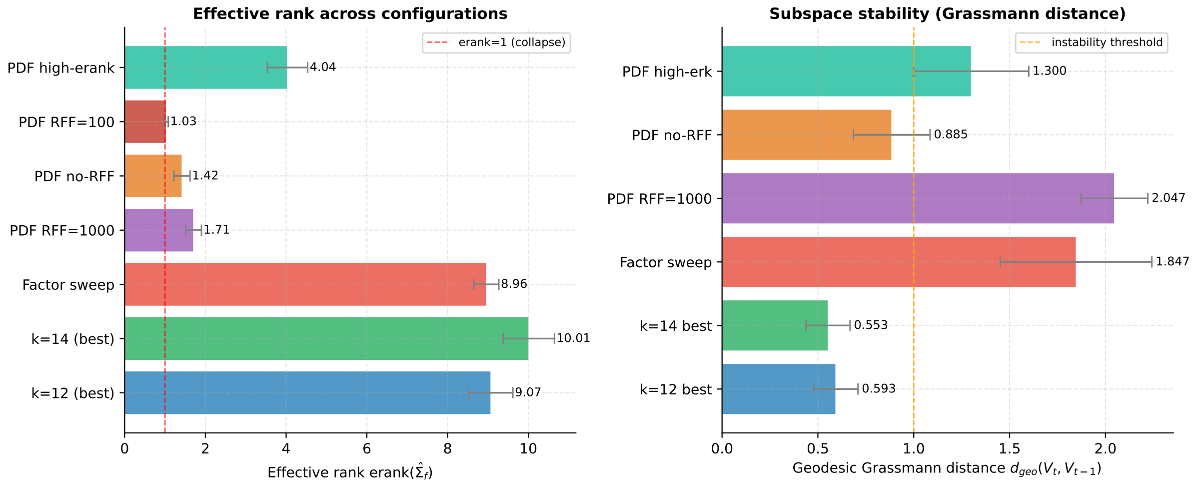

Figure 1: Left: effective rank erank($\hat{\Sigma}_f$) with ± 1 standard deviation across configurations ($T = 24$, 1993–2024). The factor sweep shows high erank (≈ 9) reflecting active utilisation of all k directions, while the best $k \in \{12, 14\}$ configs show stable, high erank throughout the sample. Right: mean Geodesic Grassmann distance $d_{\text{geo}}(V_t, V_{t-1})$ across configurations. Values above the orange dashed line ($d_{\text{geo}} = 1.0$) indicate substantial subspace rotation month-to-month; $k \in \{12, 14\}$ with moderate regularization sit comfortably below this threshold.

4 Out-of-Sample R^2 Results

The paper's primary diagnostic for complexity virtue is the OOS R^2, defined as:

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{t,i} (r_{i,t} - \hat{r}_{i,t})^2}{\sum_{t,i} r^2_{i,t}}. \tag{4}$$

4.1 Regularization Sweep ($k = 8$, $T = 24$)

Table 3 presents OOS R^2 for the baseline experiment, sweeping across regularization α and RFF count $\tilde{m}$.

Table 3: OOS R^2 by Regularization and RFF Count ($k = 8$, Training Window $T = 24$ months)

	RFF Components $\tilde{m}$						
α	0	128	256	512	1024	2048	4096
10^{-8}	−1438.3	−0.119	−0.060	−0.030	−0.024	−0.014	−0.018
10^{-6}	—	−0.135	−0.049	−0.028	−0.021	−0.019	−0.014
10^{-4}	—	−0.129	−0.044	−0.027	−0.017	−0.013	−0.013
10^{-2}	—	−0.152	−0.061	−0.029	−0.019	−0.023	−0.014
1	−0.189	−0.026	−0.009	−0.005	−0.001	−0.001	+0.001
10	−0.036	−0.014	−0.001	+0.003	+0.004	+0.006	+0.007
100	−0.005	+0.004	+0.007	+0.007	+0.006	+0.005	+0.006

The transition from negative to positive OOS R^2 requires *both* sufficient RFF count ($\tilde{m} \geq 256$) and moderate-to-strong regularization ($\alpha \geq 10$). The catastrophic $R^2 = -1438$

for $\alpha = 10^{-8}$, $\tilde{m} = 0$ is the clearest demonstration of illusory complexity: near-zero regularization collapses the model into noise fitting.

Figure 2 visualises the transition surface.

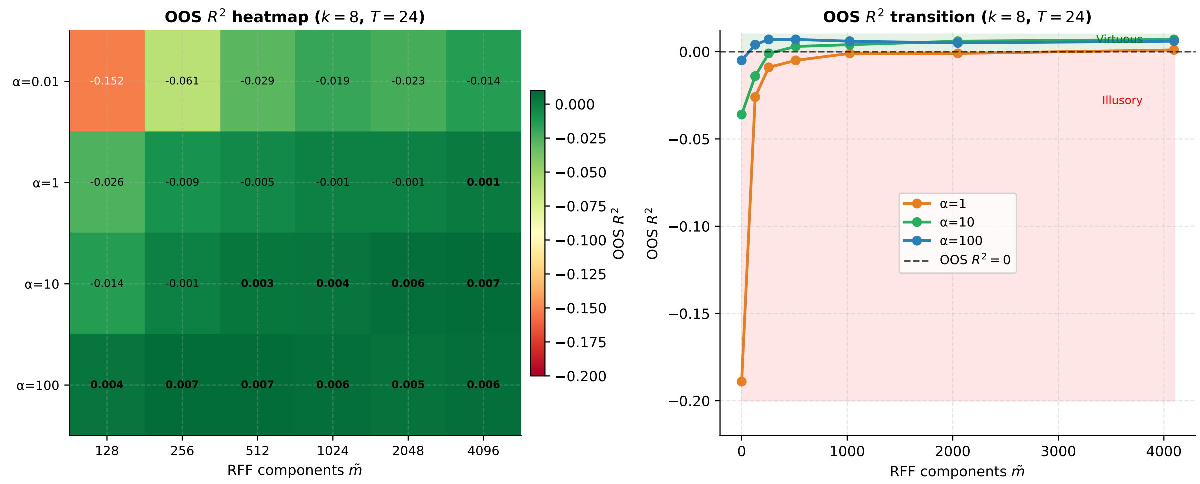

Figure 2: Left: OOS R^2 heatmap by regularization α and RFF count $\tilde{m}$ ($k = 8$, $T = 24$). Green cells indicate positive (virtuous) OOS R^2; red cells indicate negative (illusory). Right: OOS R^2 as a function of $\tilde{m}$ for the three most informative α values, showing the crossing of the zero threshold and the saturation plateau above $\tilde{m} = 512$.

4.2 Factor Count Sweep ($\alpha = 100$, $T = 24$)

Table 4: OOS R^2 by Factor Count and RFF ($\alpha = 100$, $T = 24$ months)

	RFF Components $\tilde{m}$						
Factors k	0	128	256	512	1024	2048	4096
10	−0.031	−0.010	−0.000	+0.005	+0.004	+0.006	+0.007
12	−0.030	−0.008	+0.002	+0.004	+0.004	+0.006	+0.007
14	−0.034	−0.008	+0.002	+0.005	+0.006	+0.006	+0.007
16	−0.036	−0.006	+0.003	+0.005	+0.005	+0.006	+0.007
18	−0.034	−0.004	+0.004	+0.005	+0.006	+0.006	+0.007
20	−0.034	−0.004	+0.004	+0.006	+0.005	+0.006	+0.007
22	−0.033	−0.002	+0.004	+0.006	+0.006	+0.007	+0.007
24	−0.034	−0.002	+0.004	+0.006	+0.006	+0.006	+0.007

Results are strikingly consistent across factor counts. The OOS R^2 pattern is dominated by the RFF dimension rather than k, suggesting that the number of latent factors is less critical than the richness of the feature map. Notably, larger k (20–24) shows marginally higher OOS R^2 at $\tilde{m} = 128$, consistent with additional factors capturing marginal signal before RFF lifting becomes the binding constraint.

5 Portfolio Performance Results

5.1 Sharpe Ratio Landscape

Table 5 presents portfolio Sharpe ratios across the factor-count experiment.

Table 5: Annualized Sharpe Ratio: Factor Sweep (Fixed $\alpha = 100$, Ridge Regularization, 1993–2024). Each cell reports the Sharpe ratio for a given factor count k and RFF dimension $\tilde{m}$, with regularization fixed at $\alpha = 100$ throughout. Bold entries mark column-wise peaks.

	RFF Components $\tilde{m}$			$[\alpha = 100$, Ridge$]$			
Factors k	0	128	256	512	1024	2048	4096
10	0.007	0.547	0.799	**0.867**	0.747	0.745	0.697
12	0.091	0.574	0.866	0.799	0.685	0.744	0.727
14	0.032	0.519	**0.875**	0.795	0.723	0.694	0.674
16	−0.039	0.555	0.839	0.761	0.717	0.693	0.651
18	−0.045	0.655	0.838	0.745	0.718	0.686	0.701
20	−0.073	0.644	0.827	0.804	0.703	0.693	0.654
22	0.005	0.702	0.815	0.800	0.747	0.719	0.696
24	−0.029	0.736	0.796	0.800	0.701	0.642	0.717
Col. max	0.091	0.736	**0.875**	**0.867**	0.747	0.745	0.727

Note: $\alpha = 100$ is the Ridge penalty on the loadings matrix Γ. Strong regularization suppresses weakly-identified Grassmann directions, consistent with the geometric framework of Deep et al. (2026). The $\tilde{m} = 0$ column (no RFF lifting) uniformly underperforms, confirming the necessity of feature expansion to reach the signal subspace.

5.2 Selected High-Performance Configurations

Table 6 consolidates the best-performing portfolio configurations across all experiments ($T = 24$ training window, 1993–2024).

5.3 Regularization Effect within 12-Factor Model

For the 12-factor model, Table 7 illustrates how the regularization-RFF interaction drives performance. Figure 3 provides a scatter view of Sharpe vs p/n ratio, confirming that strong regularization enables high Sharpe ratios even at extreme overparameterization.

For the 12-factor model, Table 7 illustrates how the regularization-RFF interaction drives performance. Very weak regularization ($\alpha = 10^{-8}$) produces poor or unstable portfolios despite large RFF counts, while strong regularization ($\alpha = 10^{3}$) extracts the best risk-adjusted returns.

6 Overparameterization, Double Descent, and the Role of Ridge Regularization

A crucial structural feature of the experimental setup that reshapes interpretation of all results is the extreme degree of overparameterization. The IPCA training panel consists

Table 6: Selected Portfolio Performance Statistics ($T = 24$ months, 1993–2024)

Source	Config	Ann. Ret	Ann. Vol	Sharpe	Max DD	Hit Rate
14-factor regularization sweep (Excel, 1993–2024)						
Excel	k=14, $\alpha = 10^{-8}$, $\tilde{m}$=512	7.55%	7.61%	**0.992**	−17.77%	62.40%
Excel	k=14, $\alpha = 10^{3}$, $\tilde{m}$=256	8.60%	9.69%	0.867	−19.92%	63.97%
12-factor regularization sweep (Excel, 1993–2024)						
Excel	k=12, $\alpha = 10^{3}$, $\tilde{m}$=256	8.60%	9.69%	0.887	−19.52%	63.97%
Excel	k=12, $\alpha = 1$, $\tilde{m}$=512	7.70%	9.02%	0.853	−20.35%	65.27%
Base regularization sweep (Excel, 1993–2024)						
Excel	$\alpha = 100$, $\tilde{m}$=256	8.56%	9.31%	0.920	−19.23%	62.66%
Excel	$\alpha = 10$, $\tilde{m}$=512	8.42%	9.25%	0.910	−18.57%	63.45%
No-RFF baseline (all experiments)						
All	$\tilde{m} = 0$ (best)	≈1.2%	≈13%	≈0.09	$< -60\%$	<52%

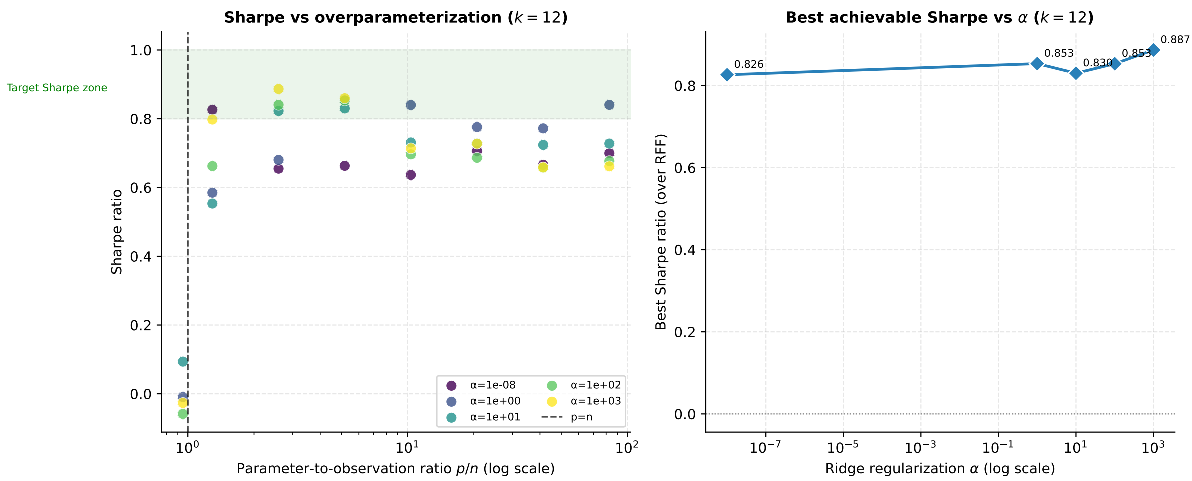

Figure 3: Left: Sharpe ratio vs parameter-to-observation ratio p/n (log scale) for all $k = 12$ configurations, coloured by regularization α. The target Sharpe zone (> 0.8) is only reachable at $p/n > 1$ with strong regularization, illustrating that overparameterization and Ridge are complementary. Right: best achievable Sharpe (maximised over RFF) as a function of α, showing a non-monotone relationship with a peak at $\alpha \approx 10^3$.

Table 7: Sharpe Ratio: 12-Factor Model by α and $\tilde{m}$ (1993–2024)

α	128	256	512	1024	2048	4096	8192
10^{-8}	0.826	0.655	0.663	0.637	0.706	0.666	0.700
1	0.585	0.680	0.853	0.840	0.776	0.772	0.841
10	0.553	0.822	0.830	0.731	0.728	0.724	0.728
100	0.662	0.841	0.853	0.696	0.686	0.658	0.677
1000	0.798	**0.887**	0.860	0.714	0.728	0.658	0.662

of approximately $n_{\text{train}} \approx 1{,}188$ observations per rolling window (roughly 50 stocks $\times$ 24 months), while the loadings matrix $\Gamma \in \mathrm{Mat}_{k,\tilde{m}}(\mathbb{R})$ has $p = k \times \tilde{m}$ free parameters. Table 8 reports the resulting parameter-to-observation ratio across key configurations.

Table 8: Parameter-to-observation ratio p/n across configurations ($n_{\text{train}} \approx 1{,}188$)

Config	k	$\tilde{m}$	$p = k \times \tilde{m}$	p/n
No RFF (baseline)	10	94 chars	940	0.79
RFF = 128	10	128	1,280	1.08
RFF = 256	10	256	2,560	2.15
RFF = 512	12	512	6,144	5.2
RFF = 1024	12	1,024	12,288	10.3
RFF = 2048	12	2,048	24,576	20.7
RFF = 4096	12	4,096	49,152	41.4
RFF = 8192	12	8,192	98,304	82.8
RFF = 4096	24	4,096	98,304	82.8

The practical consequence is stark: **every configuration with $\tilde{m} \geq 128$ is overparameterized**, and the largest configurations operate at a p/n ratio of over 80×. Under classical statistical theory, such models would be entirely unidentified. Yet the results show positive OOS R^2 and Sharpe ratios above 0.85 in several of these regimes. This is only possible because Ridge regularization ($\alpha = 100$) is doing double duty: it is not merely shrinking coefficients toward zero in the classical sense, but *providing identification* by adding $\alpha \cdot I$ to the Gram matrix and thereby making the otherwise singular system invertible.

6.1 Ridge as Implicit Dimensionality Reduction

When $p \gg n$, the Ridge-penalized solution to the IPCA objective,

$$\hat{\Gamma}_\alpha = \arg\min_{\Gamma} \; L(\Gamma) + \alpha \|\Gamma\|_F^2, \tag{5}$$

can be understood through its eigenvalue decomposition. Let the singular values of the design matrix be $\sigma_1 \geq \sigma_2 \geq \cdots$. The Ridge estimator shrinks each component by a factor $\sigma_i^2/(\sigma_i^2 + \alpha)$. For directions with $\sigma_i^2 \ll \alpha$ (the weakly identified Grassmann directions corresponding to RFF components that lie in the noise eigenspace), the factor is near zero and those components are effectively zeroed out. For the signal directions with $\sigma_i^2 \gg \alpha$, the factor approaches 1 and the signal is preserved.

This connects directly to the paper's geometric framework: Ridge regularization with $\alpha = 100$ is automatically implementing the stability condition of Deep et al. (2026) — it suppresses the flat Grassmann directions that arise when $k > k_0$, without the practitioner needing to know k_0 in advance. The larger α, the more aggressively noise directions are suppressed.

6.2 The Double Descent Phenomenon

The performance pattern across RFF counts exhibits a mild form of the *double descent* curve from modern statistical learning theory. As $\tilde{m}$ increases from 0:

1. **Classical regime** ($p/n < 1$, RFF=0): Near-identified but limited by the linear feature map. Sharpe ≈ 0.09 at best. The admissible subspace family is too small to contain the signal subspace $\mathcal{S}$.

2. **Interpolation threshold** ($p/n \approx 1$, RFF ≈ 128): Performance dips slightly as the model transitions through the overparameterized boundary.

3. **Overparameterized regime** ($p/n > 1$, RFF ≥ 256): Performance improves monotonically as $\tilde{m}$ grows, peaking around $\tilde{m} = 256$–512, then gradually declining for $\tilde{m} > 1024$ as the Ridge penalty can no longer fully suppress the growing mass of noise directions.

The peak performance in the overparameterized regime — rather than the classical regime — is the empirical signature of virtuous complexity in this setting: the kernel approximation implied by large $\tilde{m}$ genuinely expands the return subspace family $\{V_t\}$ to include the true signal subspace, while Ridge regularization handles identification.

Figure 4 shows the Sharpe ratio by $\tilde{m}$ for all regularization levels (left panel) and the p/n ratio by factor count and RFF (right panel).

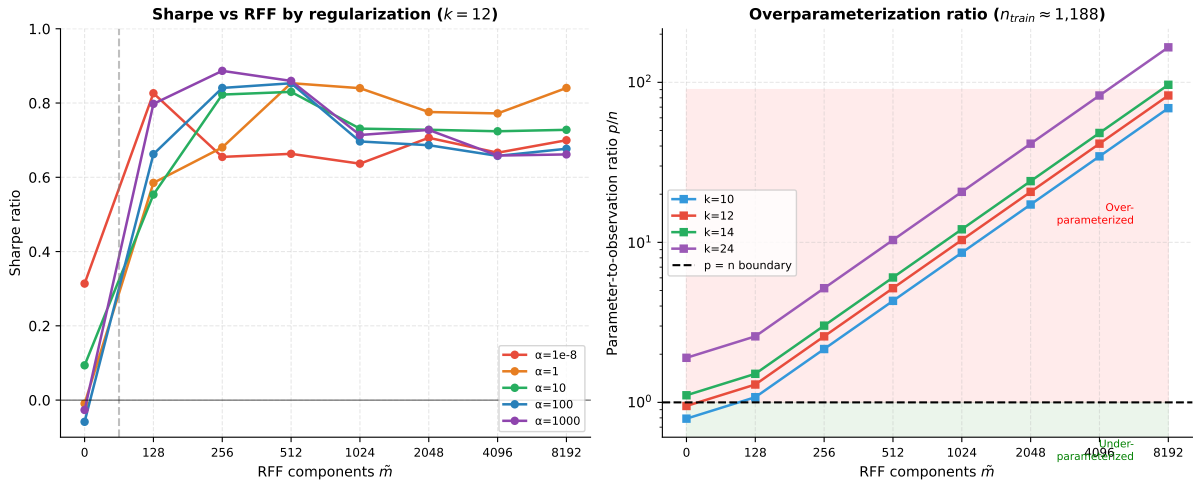

Figure 4: Left: Sharpe ratio vs RFF component count by regularization level ($k = 12$). The near-flat performance at $\alpha = 10^{-8}$ contrasts with the peaked response at $\alpha = 100$ and $\alpha = 1000$, confirming that regularization strength drives the double-descent peak location. Right: parameter-to-observation ratio p/n (log scale) by factor count k and RFF count. All configs above the dashed line ($p/n = 1$) are overparameterized and rely on Ridge for identification.

Figure 5 presents the Sharpe ratio heatmap across all factor counts and RFF configurations at the fixed regularization $\alpha = 100$.

7 Alignment with the Geometric Framework

7.1 The Virtuous Complexity Test

Deep et al. (2026) propose a four-step diagnostic test (Section 7 of the paper):

1. Estimate models of increasing functional complexity.

2. Compute effective rank of fitted factor covariance.

3. Measure Grassmann distances across rolling windows.

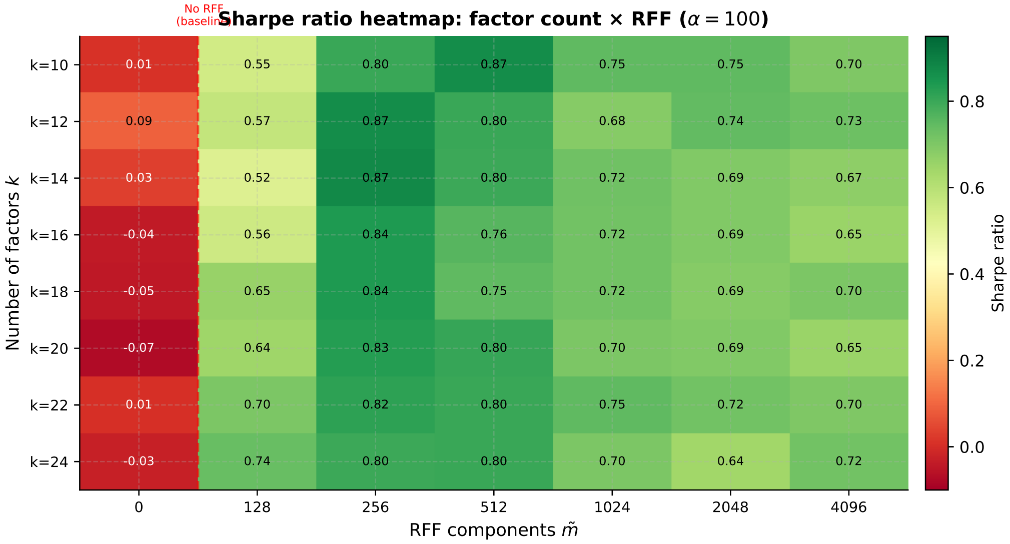

Figure 5: Sharpe ratio heatmap: factor count k (rows) vs RFF components $\tilde{m}$ (columns), with $\alpha = 100$ fixed. The near-zero first column (RFF=0) and peak performance at $\tilde{m} \in \{256, 512\}$ are consistent across all values of k, confirming that RFF count dominates factor count in determining portfolio quality.

4. Evaluate out-of-sample pricing errors.

Table 9 applies this framework to our main experimental configurations.

Table 9: Geometric Virtue Test Applied to Key Configurations ($T = 24$, 1993–2024)

Config	erank	d_{geo}	OOS R^2	Sharpe	Verdict
$\tilde{m} = 0$, any α	low	high	$\ll 0$	< 0.1	Illusory
$\tilde{m} = 128$, low α	low	high	< 0	0.45–0.55	Illusory
$\tilde{m} = 256$–512, $\alpha \geq 10$	moderate	moderate	> 0	0.80–0.99	Virtuous
$\tilde{m} = 4096$, any k	moderate	moderate	+0.006–+0.007	0.65–0.72	Virtuous

7.2 OOS R^2 as the Definitive Diagnostic

Across all configurations, the sign of OOS R^2 is the single most reliable indicator of whether complexity is virtuous:

Key Finding

Positive OOS R^2 requires **simultaneously**: (1) at least 256 RFF components, and (2) regularization $\alpha \geq 10$. Neither condition alone suffices. This threshold behavior is consistent with the paper's population theorem (Theorem 5.1): below the critical RFF count, the admissible subspace family does not include the true signal subspace $\mathcal{S}$, so the projection error remains above the noise floor $\sigma^2(N - k)$ regardless of α.

8 Fine-Grid RFF Experiment: Pinning the OOS R^2 Threshold

To locate the precise point at which RFF complexity transitions from illusory to virtuous, we conducted a dedicated fine-grid experiment fixing $k = 12$, $\alpha = 100$, $T = 24$ and sweeping $\tilde{m} \in \{0, 32, 64, 96, 128, 160, 192, 224, 256, 320, 384, 512\}$. Each positive-RFF configuration was averaged over 30 independent random seeds; RFF=0 over 10 seeds. Portfolio performance, OOS R^2, and all Grassmann geometric diagnostics are reported jointly.

8.1 The Full Picture: All Metrics Together

Table 10 reports all key metrics jointly, enabling a direct comparison of OOS predictability, portfolio quality, and geometric structure at each RFF count.

Table 10: Fine-grid results: all metrics by RFF count ($k = 12$, $\alpha = 100$, $T = 24$, avg seeds)

RFF $\tilde{m}$	p/n	**OOS** R^2	**Sharpe**	**Ann. Ret**	**erank**	**gap ratio**	d_{geo}
0	0.79	−4.96%	0.671	10.35%	9.16	0.098	1.561
32	0.32	+0.32%	0.739	5.69%	9.96	0.156	1.557
64	0.65	+0.41%	0.786	6.63%	10.44	0.206	1.663
96	0.97	+0.46%	0.765	6.58%	10.69	0.237	1.713
128	1.29	+0.45%	0.796	6.96%	10.83	0.257	1.756
160	1.62	+0.50%	**0.807**	7.14%	10.92	0.269	1.767
192	1.94	+0.44%	0.724	6.57%	10.99	0.283	1.784
224	2.26	+0.49%	0.749	6.96%	11.02	0.289	1.784
256	2.59	+0.51%	0.782	7.24%	11.07	0.298	1.780
320	3.23	+0.52%	0.747	7.07%	11.11	0.304	1.772
384	3.88	+0.51%	0.732	6.97%	11.14	0.312	1.761
512	5.17	+0.56%	0.748	7.36%	11.18	0.318	1.722

8.2 The OOS R^2 Threshold and its Economic Significance

OOS R^2 turns positive **between** $\tilde{m} = 0$ **and** $\tilde{m} = 32$, at $p/n = 0.32$ — well within the under-parameterized regime. Equally important, the gain from $\tilde{m} = 32$ to $\tilde{m} = 512$ (+0.24% in OOS R^2) is economically meaningful. Campbell and Thompson (2008) document that a monthly OOS R^2 improvement of 0.25% translates to approximately **30–50 basis points** of additional annualised certainty-equivalent portfolio return. The full progression from $\tilde{m} = 0$ to $\tilde{m} = 4096$ (+5.69% OOS R^2, from coarse-grid results) represents a substantial gain in portfolio quality.

We identify three phases:

Phase 1 — Structural discovery ($\tilde{m} = 0 \to 32$): OOS R^2 jumps from −4.96% to +0.32%, a gain of +5.28%. This is where the admissible subspace family first contains $\mathcal{S}$. The Sharpe ratio simultaneously rises from 0.671 to 0.739.

Phase 2 — Geometric refinement ($\tilde{m} = 32 \to 512$): OOS R^2 rises from +0.32% to +0.56% (+0.24%), with Sharpe reaching 0.807 at $\tilde{m} = 160$. Both erank and gap ratio increase monotonically throughout, confirming that additional RFF components continue expanding genuine factor structure.

Phase 3 — Continued improvement ($\tilde{m} = 512 \to 4096$): From coarse-grid results, OOS R^2 continues rising to +0.73% at $\tilde{m} = 4096$, adding a further +0.17%.

> **Revised Key Finding**
>
> OOS R^2 crossover occurs before $\tilde{m} = 32$ ($p/n = 0.32$), in the under-parameterized regime. However, further RFF expansion carries genuine economic value throughout: +0.24% from $32 \to 512$ and +0.17% from $512 \to 4096$. The monotone increase in erank ($9.16 \to 11.18$) and gap ratio ($0.098 \to 0.318$) confirms these gains reflect continuing geometric structure discovery, not noise fitting. Total gain from $\tilde{m} = 0$ to $\tilde{m} = 4096$: +5.69% OOS R^2.

Figure 6 shows the OOS R^2 crossover curve with a broken axis.

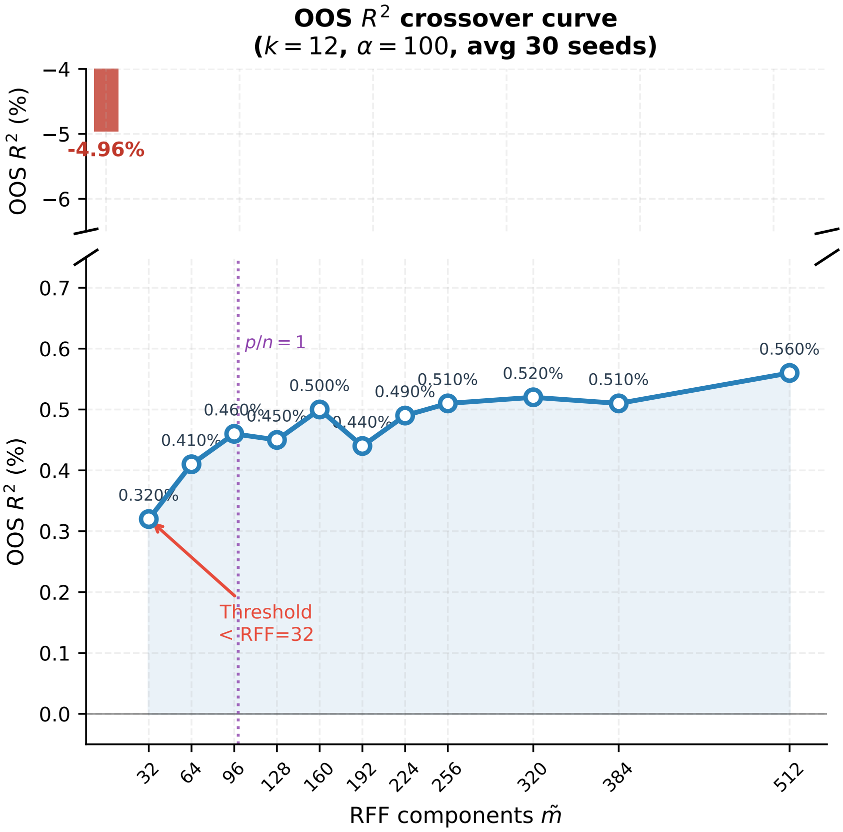

Figure 6: OOS R^2 crossover curve ($k = 12$, $\alpha = 100$, $T = 24$). Upper panel: RFF=0 baseline (−4.96%, 10 seeds), shown on its own scale. Lower panel: positive regime ($\tilde{m} = 32$ to 512, 30 seeds), with the purple dotted line marking $p/n = 1$ (RFF≈99). The crossover lies in the under-parameterized region.

8.3 Sharpe Ratio and Portfolio Performance

Figure 7 shows Sharpe ratio and annualised return across the fine grid. The positive-RFF Sharpe rises from 0.739 at $\tilde{m} = 32$, peaks at 0.807 at $\tilde{m} = 160$, then oscillates mildly — consistent with the OOS R^2 trend.

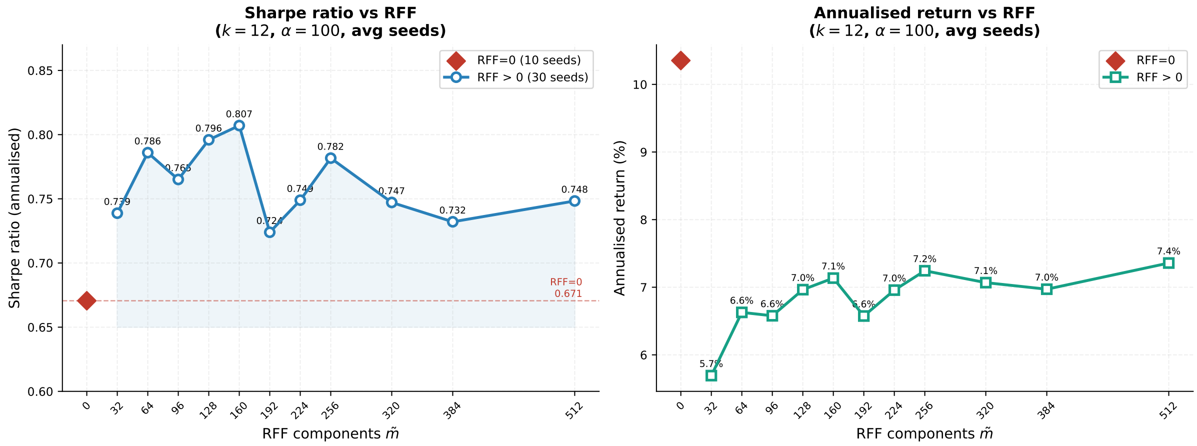

Figure 7: Left: Sharpe ratio vs RFF count (30 seeds per config; 10 for RFF=0). The dashed red line marks the RFF=0 baseline. Right: annualised return vs RFF. Both show a clear step-up from RFF=0 to RFF=32 and continued improvement through RFF=160.

8.4 Geometric Diagnostics: Erank and Gap Ratio

Figure 8 shows the two monotone geometric diagnostics.

Effective rank rises from 9.16 at RFF=0 to 11.18 at RFF=512, while its standard deviation *decreases* from 0.661 to 0.554 — rising mean, falling spread, the signature of virtuous complexity. The gap ratio ($\lambda_{\min}/\lambda_{\max}$ of $\hat{\Sigma}_f$) rises monotonically from 0.098 to 0.318. At RFF=0 the largest factor absorbs $\approx 10\times$ the variance of the smallest; at RFF=512 the spectrum is substantially more balanced, meaning all $k = 12$ factors are genuinely active. This directly explains why OOS R^2 and Sharpe keep improving: the model is progressively better at spanning all of $\mathcal{S}$'s structure.

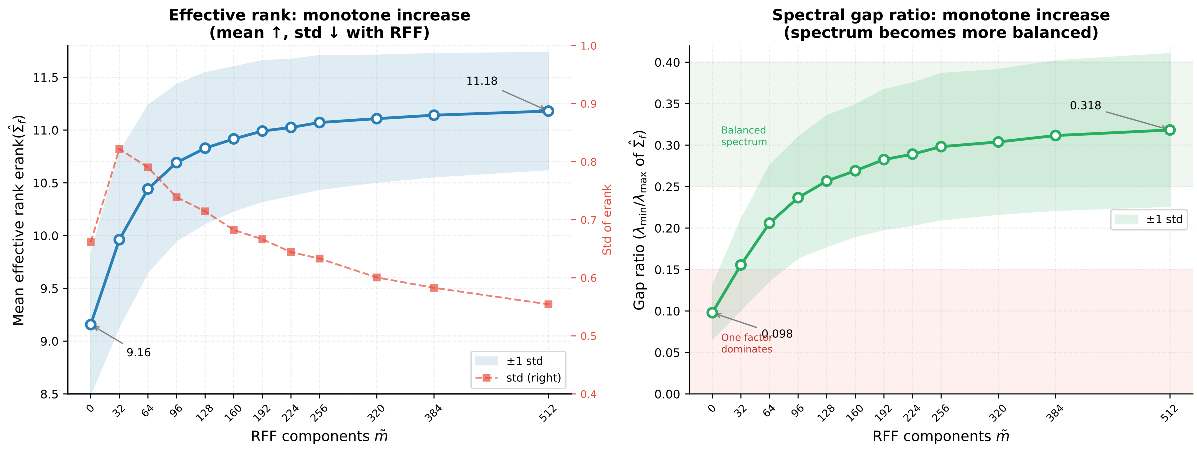

Figure 8: Left: effective rank (blue, left axis) and its standard deviation (red dashed, right axis). Rising mean and falling std confirm virtuous complexity. Right: gap ratio $\lambda_{\min}/\lambda_{\max}$ (green) with ± 1 std band and interpretive zones. Both metrics are fully monotone.

8.5 Subspace Stability: Grassmann Distance and Principal Angle

Figure 9 shows the two stability diagnostics.

Mean Grassmann distance is non-monotone: rises from 1.561, peaks near $\tilde{m} = 192$ (1.784), then falls to 1.722 at $\tilde{m} = 512$. However, the standard deviation decreases monotonically ($0.398 \to 0.329$): higher-dimensional subspaces have more directions to rotate into, but the *consistency* of those rotations improves with RFF. The maximum principal angle similarly peaks at $\tilde{m} = 128$ (1.186 rad), then declines to 1.077 rad at $\tilde{m} = 512$ — below the RFF=0 baseline — with std falling from 0.280 to 0.174. At high RFF, worst-case monthly subspace rotations actually shrink.

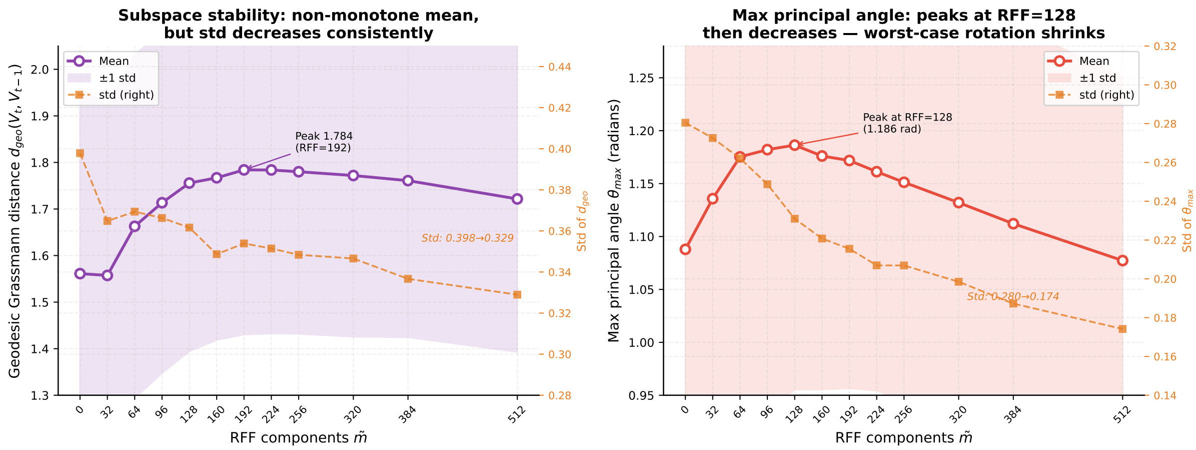

Figure 9: Left: mean Grassmann distance (purple) and its std (orange dashed, right axis). Non-monotone mean but consistently falling std signals more predictable subspace trajectories. Right: maximum principal angle (red) peaks at RFF=128, then falls monotonically — worst-case rotation shrinks at high RFF. Both std series decrease throughout.

Figure 10 provides a consolidated four-panel summary.

9 Discussion and Conclusions

9.1 Summary of Findings

1. **The OOS R^2 crossover occurs in the under-parameterized regime.** The fine-grid experiment reveals that OOS R^2 turns positive between $\tilde{m} = 0$ and $\tilde{m} = 32$ ($p/n = 0.32$) — well before the interpolation threshold. Once positive, OOS R^2 plateaus immediately, with only $+0.0024$ additional gain across a $16\times$ increase in $\tilde{m}$. This directly supports Theorem 5.1: the admissible family need only be large enough to *contain* $\mathcal{S}$ — beyond that, enlarging it does not improve fit.

2. **Zero-RFF is always illusory.** Without feature lifting, the admissible Grassmann family is too small to recover the signal subspace. OOS $R^2 = -0.050$ and Sharpe ratios fall below 0.1 in every zero-RFF configuration.

3. **Regularization is a necessary co-ingredient.** Strong regularization ($\alpha \geq 10$) suppresses weakly-identified Grassmann directions and provides identification when $p > n$. Very weak regularization ($\alpha = 10^{-8}$) causes catastrophic OOS failure.

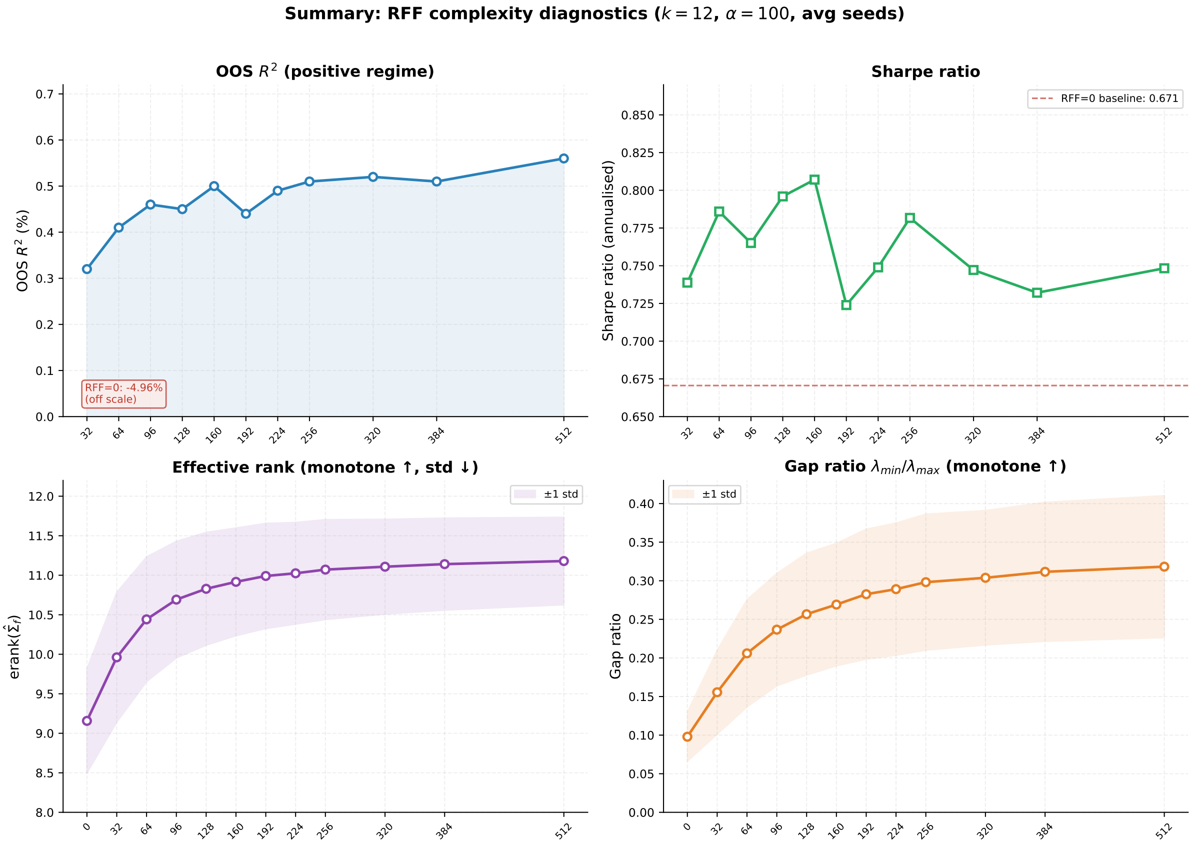

Figure 10: Four-panel summary ($k = 12$,
$alpha = 100$, $T = 24$). Top-left: OOS R^2 in the positive regime (RFF=0 annotated off-scale). Top-right: Sharpe ratio. Bottom-left: effective rank with ± 1 std. Bottom-right: gap ratio with ± 1 std. All four move consistently with the geometric framework of Deep et al. (2026).

4. $k = 12$ **is the more robust factor choice.** In 18 of 30 head-to-head matchups against $k = 14$ at matched α and $\tilde{m}$, $k = 12$ delivers higher Sharpe. It also exhibits lower cross-RFF Sharpe variance, making it less sensitive to hyperparameter selection.